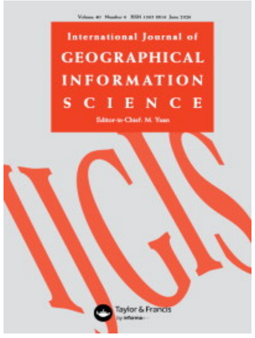

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RESEARCH ARTICLE

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Adaptive dynamic graph learning for forecasting urban multimodal flow

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ABSTRACT

The increasing diversity and integration of transportation modes is changing urban mobility, resulting in complex spatiotemporal urban flow patterns. The current forecasting models, which typically rely on static or manually defined graph structures, are inadequate for capturing the dynamic spatial heterogeneity and complex cross-modal interactions that are present in real urban systems. To address these limitations, this study introduces a multimodal dynamic graph neural network (MM-DyGNN), a novel deep learning model that is designed for urban multimodal flow prediction. MM-DyGNN introduces three key innovations: (i) a time-varying multi-modal graph learning module based on Tucker decomposition that adaptively constructs mode- and time-specific diffusion graphs; (ii) a sparse cross-modal interaction module that employs a top-*k* strategy to capture the most relevant region–mode dependencies; and (iii) an adaptive multitask learning strategy with uncertainty weighting to balance heterogeneous modal objectives. Comprehensive experiments conducted on real-world urban mobility datasets demonstrate that the MM-DyGNN significantly outperforms the baseline models in terms of forecasting accuracy. Ablation studies further validate the effectiveness of each component and demonstrate the ability of the model to interpret dynamic spatiotemporal dependencies and cross-modal interactions. This work provides a methodological foundation for understanding and managing the evolving complexities of urban mobility.

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1. Introduction

With rapid urbanization, traffic congestion, energy consumption, and environmental pollution have become major obstacles to sustainable urban development (Yin *et al.* 2024). To address these challenges, smart cities and intelligent transportation systems (ITSs) have been developed. These systems aim to optimize traffic management,

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improve operational efficiency, and improve the travel experience through advanced information and communication technologies (Li *et al.* 2024). A pivotal technology within ITSs is accurate and real-time urban flow forecasting, which refers to predicting the volume of trips departing from or arriving at a spatial unit within a given time interval. As urban transport infrastructure improves, citizens increasingly exhibit multimodal travel behaviors, including the use of metro systems, buses, taxis, and shared bicycles. Consequently, forecasting urban multimodal flow, as opposed to single-mode usage, is essential. Accurate urban multimodal flow forecasting enables smarter traffic planning schemes and better travel services by optimizing the distribution of resources between different modes of transportation. (Tao *et al.* 2023).

In recent years, deep learning has driven remarkable progress in traffic prediction. Recurrent neural networks (RNNs) have been widely applied to capture temporal dependencies in sequential traffic data (Cui *et al.* 2024). To capture spatial dependencies, convolutional neural networks (CNNs) and, more recently, graph neural networks (GNNs) have been employed to represent urban transportation systems. Building on these advances, spatiotemporal GNNs such as STGCN (Yu *et al.* 2018), ASTGCN (Guo *et al.* 2019), and Graph WaveNet (Wu *et al.* 2019) have demonstrated strong performance in single-mode traffic prediction. These methods have demonstrated strong performance in single-mode transport prediction scenarios. With respect to multimodal transport prediction, the existing approaches often treat different transportation modes (e.g., metros, buses, and taxis) as multiple input channels or apply feature fusion strategies to unify heterogeneous inputs. Some recent studies have proposed modality-sensitive graph structures or cross-modal attention mechanisms to achieve improved integration performance (Liang *et al.* 2022b, Wang *et al.* 2022).

However, extending these approaches to urban multimodal flow forecasting introduces several unique challenges. The **first challenge** encountered when predicting urban multimodal flow lies in the dynamic spatial heterogeneity that is present both within and across transportation modes. Although GNN-based models such as the STGCN and ASTGCN have shown strong performance in terms of forecasting urban flow (Guo *et al.* 2019), their core assumption involves the capture of spatial dependencies through a preconstructed static graph structure. Their adjacency matrices are often defined on the basis of prior knowledge, such as the geographical distance, road connectivity, or functional similarity between regions (Huang *et al.* 2022a). However, real-world urban traffic systems exhibit dynamic and time-varying structures that are not adequately represented by static graphs. As illustrated in Figure 1 (a), intramodal spatial correlations vary over time. This means that a fixed, static graph cannot accurately capture the true intramodal dynamics of urban flow, as the underlying relationships between regions are constantly shifting.

The **second challenge** lies in the intricate spatial interactions among different transport modes. The flow patterns of various modes, such as metros, buses, and taxis, do not evolve independently but rather exhibit significant mutual influences (Zhao *et al.* 2023). These intermodal dependencies are shaped by shared infrastructure, overlapping service areas, and the competitive or complementary relationships between modes (Zhang *et al.* 2023). However, not all intermodal interactions are equally informative. Long-range or semantically unrelated connections can introduce significant noise

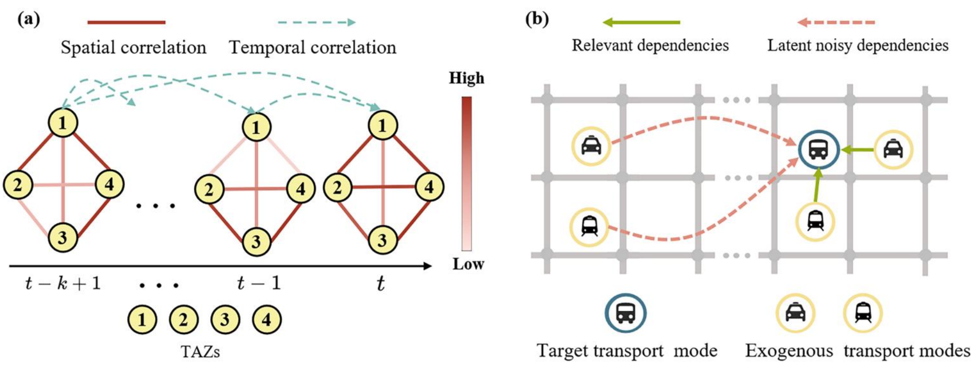

Figure 1. Key concepts in multimodal urban flow prediction. (a) Intramodal dynamics: temporal evolution of spatial dependencies across TAZs represented as dynamic graphs. (b) Intermodal interactions: cross-influences between transport modes (e.g., buses, metros, taxis), where distant links may indicate noise rather than meaningful connections.

and hinder the performance of the developed model. As illustrated in Figure 1(b), effective multimodal transportation system modeling necessitates the capacity to distinguish truly relevant intermodal dependencies from spurious or irrelevant associations. Therefore, it is essential for prediction models to incorporate adaptive mechanisms that selectively emphasize meaningful cross-modal correlations while attenuating the influences of noisy or uninformative links.

The **third challenge** faced when predicting urban multimodal flow lies in multitask learning. In many of the existing approaches, a common practice is to apply a uniform loss weighting scheme, i.e., calculating the overall loss by directly adding or averaging the individual losses induced by each subtask (Gong *et al.* 2019). While simple and computationally efficient, this uniform treatment fails to account for substantial task difficulty or gradient magnitude differences, as well as the potential for conflicting task objectives. Consequently, the optimization process may become biased toward subtasks that are inherently easier to learn or those that yield larger losses, thereby overshadowing harder or more nuanced tasks. This imbalance can lead to suboptimal generalization capabilities and degraded performance across underrepresented or more complex prediction targets.

To address the aforementioned challenges, a novel deep learning model named the multimodal dynamic GNN (**MM-DyGNN**) is proposed in this study to achieve more accurate and robust urban multimodal flow forecasting. We developed a multimodal time-varying dynamic graph (MTDG) learning module to address static graphs' inability to capture the dynamic, varied nature of multimodal transportation. Instead of static adjacency matrices, this module adaptively creates a dynamic traffic diffusion graph for each mode (e.g., metros, buses, taxis) at each time step. To address the issues related to noisy global dependencies in cross-modal interaction modeling tasks, we introduce a sparse cross-modal interaction (SCMI) module. This module employs a dynamic selection mechanism (e.g., a top-*k* strategy) to identify and focus on the most critical and strongly interacting local region-mode pairs in the spatial dimension. Finally, an adaptive multitask learning strategy is designed to dynamically balance multiple subtasks during training, optimizing the overall prediction performance.

The remainder of this paper is organized as follows. Section 2 reviews the related work. Section 3 specifies notation and problem setup. Section 4 presents the proposed model architecture and its key components in detail. Section 5 introduces the experimental setup, including the datasets used, the evaluation metrics, and the baseline methods, as well as a comprehensive analysis of the results. Section 6 discusses the robustness of the proposed model and the impact of hyperparameter settings on its performance. Finally, Section 7 concludes the paper.

2. Related work

2.1. Urban multimodal flow prediction

With the diversification of travel modes, urban transportation systems have become increasingly complex and dynamic. Early studies mainly focused on single-mode forecasting (e.g., buses or metros) using statistical models such as ARIMA and SVR. The advent of deep learning significantly improved prediction accuracy and generalization. RNNs and LSTMs effectively model temporal dependencies (Zhao *et al.* 2022), whereas CNNs capture spatial features. However, with emerging modes such as bike sharing and ride-hailing, single-mode prediction has become inadequate for fine-grained management. Recent studies thus emphasize multimodal flow prediction, which jointly models the coevolution of multiple modes (e.g., metro, bus, taxi, and bike) (Liang *et al.* 2022b). Existing methods typically adopt either multichannel input or multitask learning (MTL) frameworks to unify multimodal patterns (Li *et al.* 2021a, Yang *et al.* 2025, Cao *et al.* 2025b), yet they still face challenges in model design and interaction representation.

Current multimodal flow prediction approaches can be grouped into three categories: multichannel input models, MTL frameworks, and GNN-based fusion methods. Multichannel models process different modes in parallel with CNN or ConvLSTM architectures, but fail to capture explicit intermodal dependencies. MTL frameworks jointly optimize different modes (Liang *et al.* 2022a, Liu *et al.* 2021), yet rely on static loss weighting and struggle with task imbalance. GNN-based approaches such as STGCN, ASTGCN, and Graph WaveNet (Gammelli *et al.* 2021) have shown success in single-mode prediction and are extended to multimodal settings via multigraph or hypergraph structures (Ma *et al.* 2022, Cao *et al.* 2025a). However, most still depend on static graphs, ignoring temporal variations and fine-grained spatial heterogeneity. Consequently, current methods often neglect the spatiotemporal coupling among modes (e.g., simultaneous metro–bus surges) and rely on global or fully connected mechanisms that introduce redundant or noisy dependencies, limiting interpretability and generalization.

2.2. Spatial dependence modeling

Spatial dependence is a key property of urban transportation systems. Early studies adopted grid-based CNN or ConvLSTM frameworks to model spatiotemporal traffic patterns, such as the DeepST model (Guo *et al.* 2021), which combined CNNs and RNNs to capture spatial and temporal dynamics. However, grid-based methods fail to represent the non-Euclidean topology of road networks. To overcome this, GNNs were introduced to represent cities as graphs, where nodes denote regions or sensors and edges describe spatial relationships. Representative models such as STGCN (Yu *et al.* 2018), ASTGCN, and Graph WaveNet (Cao *et al.* 2023) integrate temporal convolution, attention, or diffusion mechanisms to capture spatiotemporal dependencies. Although these GNN-based models have become the mainstream paradigm for spatial modeling, they still rely on static adjacency matrices, which cannot capture dynamic spatial influence changes across time.

Recent advances have extended spatial modeling into three categories: static, dynamic, and adaptive graph learning. Static models (e.g., STGCN, ASTGCN, DCRNN) use fixed adjacency matrices derived from geography or road connectivity (Dai *et al.* 2020), which ensures interpretability but lacks temporal adaptability. Dynamic models, such as DGCRN (Li *et al.* 2021b), update adjacency matrices through gating mechanisms to reflect evolving spatial patterns (Hu *et al.* 2022). Adaptive approaches like MTGNN (Wang *et al.* 2024) directly learn topologies from multivariate time series, improving flexibility (Jiang *et al.* 2022). Nevertheless, most DGNNs focus on single-mode systems and overlook multimodal heterogeneity. They fail to model dynamic diffusion ranges—e.g., metro influence expands during peak hours but contracts off-peak (Yang *et al.* 2019)—and lack mechanisms to jointly capture intramodal and intermodal structural evolution (Dai *et al.* 2024). To address these issues, we propose an MTDG module based on Tucker decomposition, which constructs a dynamic graph tensor for multimodal systems to jointly model temporal dynamics and cross-modal heterogeneity, providing a more expressive and adaptive framework for urban transportation modeling (Mi *et al.* 2023).

2.3. Cross-modal transport interaction

With the growth of shared mobility services, urban travel behaviors have become increasingly multimodal, making coordination among transport modes a key research focus. Intermodal relationships such as complementarity, substitution, and collaboration frequently occur—for instance, rising metro ridership during peak hours often coincides with increased bus and bike flows, whereas at night, taxi services compensate for metro closures (Jin *et al.* 2022). Recent studies have begun to jointly model these interactions. Liang *et al.* (2022b) proposed ST-MRGNN to predict metro–ride-hailing flows via multirelational attention, while Yang *et al.* (2025) introduced GRAPE, integrating bike and taxi data through transmodality and equity graphs to enhance predictive accuracy and fairness. These works confirm the feasibility and importance of capturing intermodal dependencies.

Existing modeling approaches can be grouped into three categories: multichannel fusion, graph-based modeling, and attention-driven frameworks. Multichannel methods (Li *et al.* 2022) process modes as independent input channels but fail to capture strong interdependencies. Graph-based approaches, such as heterogeneous graphs in Xu *et al.* (2022), explicitly encode multimodal topologies but rely on manually defined edge weights, risking biased interaction intensity estimation. Attention-based methods (Zhang *et al.* 2024) improve flexibility by capturing both local and global dependencies, yet fully connected attention often introduces noisy or redundant relations in large-scale systems. Overall, current methods face limitations in identifying locally meaningful and temporally dynamic interactions (Wu *et al.* 2024). To overcome these challenges, we propose an SCMI mechanism that employs a dynamic top-k strategy to identify the most influential region–modality pairs in both spatial and modal dimensions. This sparse and adaptive design enables efficient structure learning, filters out redundant dependencies, and enhances model interpretability and generalization by dynamically emphasizing significant cross-modal correlations.

3. Preliminaries

The forecasting of multimodal urban flow involves specifying (i) *where* urban flow is measured (spatial regions $\mathcal{R} = \{r_1, \ldots, r_N\}$); (ii) *what* attributes are observed (the historical urban flow across modes $\mathcal{M} = \{\text{bus}, \text{taxi}, \text{metro}\}$); and (iii) *how* modes interact over a historical window T and the forecasting horizon τ. We summarize the main notations used throughout the paper in Table 1.

3.1. Definition 1 (traffic analysis zones)

We partition the target city into N TAZs, denoted as $\mathcal{R} = \{r_1, \ldots, r_N\}$, based on land-use and transport criteria. The TAZ is a commonly used spatial unit in urban and transportation studies, serving as the basic analytical element to model travel demand and mobility patterns (Castiglione *et al.* 2025; Wu *et al.* 2025). Throughout the paper, the terms 'TAZ', 'region', and 'node' are used interchangeably.

3.2. Definition 2 (urban multimodal flow)

In this study, the urban multimodal flow characterizes short-term trip boardings across different transportation modes, capturing the number of passengers boarding within each TAZ over successive time intervals. For each mode $m \in \mathcal{M}$, the flow is stored in a matrix $\mathbf{X}^m \in \mathbb{R}^{N \times T}$, where each entry $(\mathbf{X}^m)_{i,t}$ records the number of trips of mode m departing from region r_i at time step t. Mathematically, the urban multimodal flow tensor is thus defined as

$$\mathcal{X} = \left[\mathbf{X}^{\text{bus}} | \mathbf{X}^{\text{metro}} \ | \mathbf{X}^{\text{taxi}}\right] \in \mathbb{R}^{N \times T \times M}, \tag{1}$$

where N is the number of spatial regions, T is the length of the lookback window, and $M = |\mathcal{M}|$ is the number of transport modes.

Table 1. Summary of the key symbols and notations used in this paper.

Symbol(s)	Description
$\mathcal{R}, N$	Set of spatial regions (TAZs) and number of regions, respectively
$\mathcal{M}, M$	Set of transportation modes and number of modes, respectively
$m^*, \mathcal{M}^-$	Target transportation mode and exogenous modes, respectively
$\mathbf{X}^m, \mathcal{X}$	Historical flow matrices for mode m and multimodal flow tensor, respectively
T, τ	Historical window length and forecasting horizon, respectively
$\mathcal{G}_t^m, \mathcal{A}^m$	Dynamic intramode graph at slot t and its adjacency tensor, respectively
$\mathbf{H}^m, d$	Hidden representation for mode m and hidden dimensionality, respectively

3.3. Definition 3 (intramode dynamic graph)

An intramode dynamic graph represents the spatial dependencies within a single transport mode. Such graphs evolve over time slots within a day to reflect the *temporal variations* exhibited by intramode interactions. For each mode $m \in \mathcal{M}$, we define a dynamic graph for each daily time slot $t \in \{1, \ldots, N_t\}$ (N_t denotes the number of daily time slots): $\mathcal{G}_t^m = (\mathcal{V}, \mathcal{E}_t^m)$, where the nodes $\mathcal{V}$ represent TAZs and the edges $\mathcal{E}_t^m$ encode the interactions dynamically learned by the model. Specifically, an edge $e_{i,j,t}^m = (v_i, v_j, A_{t,i,j}^m)$ indicates a directed influence from node v_i to node v_j at time t, weighted by $A_{t,i,j}^m$. Aggregating the adjacency matrices yields the adjacency tensor: $\mathcal{A}^m = \{A_t^m\}_{t=1}^{N_t} \in \mathbb{R}^{N_t \times N \times N}$.

3.4. Definition 4 (intermodal interaction)

We define inter-modal interaction as the process of obtaining an enriched representation $\hat{\mathbf{H}}^{m^*}$ for a target mode m^* by adaptively fusing information from exogenous modes $\{\mathbf{H}^m\}_{m \in \mathcal{M}^-}$ via a learnable operator Φ_{cm}:

$$\hat{\mathbf{H}}^{m^*} = \Phi_{\text{cm}}\left(\mathbf{H}^{m^*}, \{\mathbf{H}^m\}_{m \in \mathcal{M}^-}\right), \tag{2}$$

This intermodal interaction describes region-level dependencies among different transportation modes, reflecting how the flow of one mode in a given TAZ may correlate with the flow of other modes across the city (e.g., complementary bus–metro patterns during morning peaks).

3.5. Problem statement (short-term multimodal urban flow forecasting)

Given the urban multimodal flow tensor $\mathcal{X}$, the task is to learn a predictor:

$$\mathcal{F} : (\mathcal{X}) \rightarrow (\hat{\mathbf{Y}}^1, \ldots, \hat{\mathbf{Y}}^M), \tag{3}$$

where each $\hat{\mathbf{Y}}^m \in \mathbb{R}^{\tau \times N}$ contains future flow for mode m, and the (k, i)-th entry $\hat{y}_{i,t+k}^m$ forecasts the flow in region r_i on horizon k ($k = 1, \ldots, \tau$) given data up to time t.

4. Methodology

4.1. Framework overview

To tackle the multimodal short-term outflow forecasting task, We propose MM-DyGNN (Figure 2), a multimodal spatiotemporal model that integrates dependencies both within and across transportation modes for precise urban flow prediction. This model consists of three components: (i) Intramodal Interaction: Captures mode-specific spatiotemporal patterns using dynamic graphs and temporal convolutions. (ii) Intermode Interaction: An adaptive attention module (SCMI) that sparsely fuses information from different transport modes. (iii) Adaptive Multitask Learning: An uncertainty-based loss function to automatically balance the training objectives for each mode.

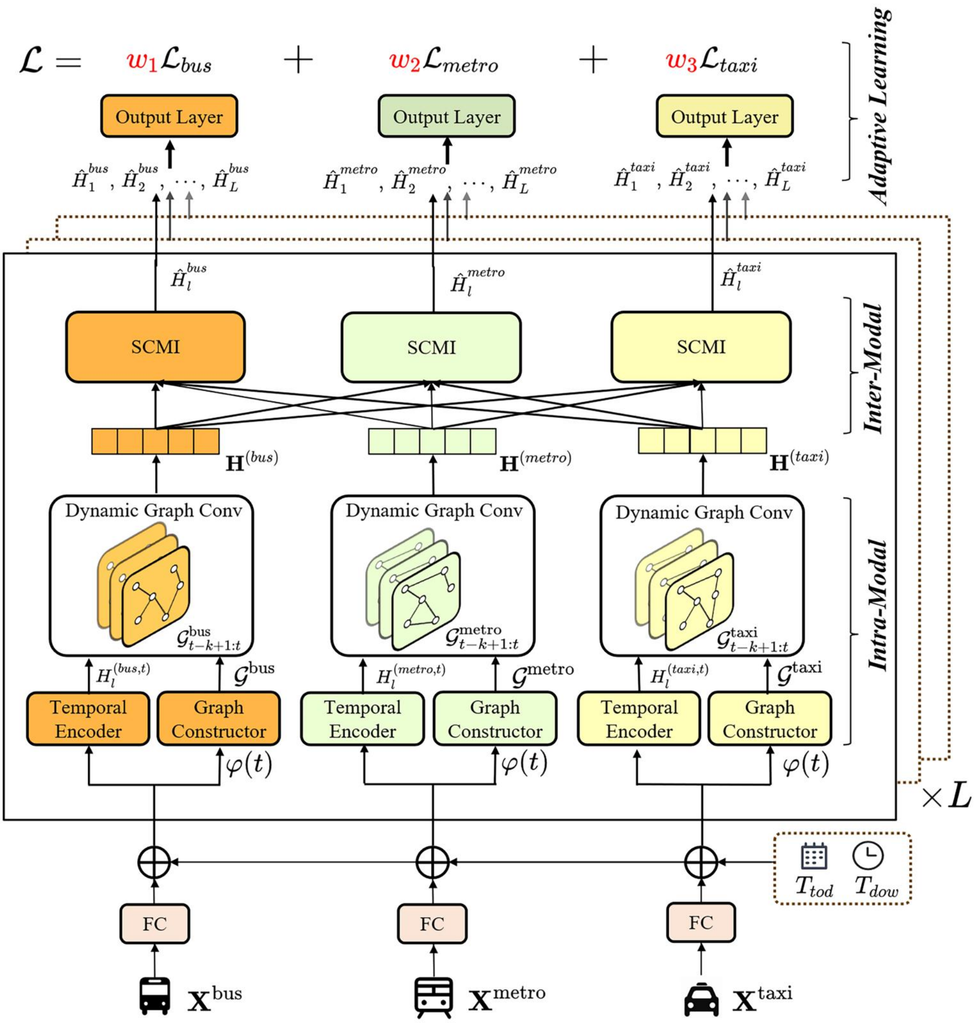

Figure 2. Overview of the proposed model. It demonstrates the intramodal interactions, intermodal interactions and adaptive learning processes among different components.

4.2. Intramodal interaction

The intramodal interaction module aims to capture fine-grained temporal and spatial dependencies within each transportation mode. It consists of three sequential stages. (i) A temporal encoder extracts long-range temporal patterns from raw flow sequences. (ii) Dynamic graph construction is performed to learn a time-varying, mode-specific adjacency matrix that reflects evolving spatial correlations. (iii) Dynamic graph convolution propagates information over the learned graphs to obtain spatially enriched representations for downstream forecasting.

4.2.1. Temporal encoder

To incorporate temporal periodicity, we first project the raw input $\mathbf{X}^m \in \mathbb{R}^{T\times N}$ for each mode m into a d-dimensional space $\mathbf{X}^m \in \mathbb{R}^{T\times N\times d}$ using a fully connected layer. We then add two learnable time embeddings: $\mathbf{T}_{tod} \in \mathbb{R}^{N_t \times d}$ (for time of day, $N_t = 144$) and $\mathbf{T}_{dow} \in \mathbb{R}^{N_w \times d}$ (for day of week, $N_w = 7$). These embeddings are broadcast to all N spatial locations, forming the enhanced representation: $\mathbf{H}^m = \mathbf{X}^m + \mathbf{X}_{tod} + \mathbf{X}_{dow}$. This output tensor $\mathbf{H}^m$, which now encodes periodicity, is passed to the temporal encoder.

To efficiently capture long-range temporal dependencies, the temporal encoder uses a stack of L temporal encoder blocks. Each block contains two dilated-causal 1-D convolutions, a Gated Linear Unit (GLU), and a 1×1 convolution:

$$\mathbf{H}_l^{(m,t)} = \mathrm{Conv}_{1\times 1}\left(\tanh\left(\mathbf{H}_{l-1}^{(m)} * W_{f,l}^{m}\right) \odot \sigma\left(\mathbf{H}_{l-1}^{(m)} * W_{g,l}^{m}\right)\right), \tag{4}$$

where $*$ is a dilated-causal convolution with a dilation factor of 2^{l-1}. The tanh and σ (sigmoid) functions form the GLU, and $\odot$ denotes element-wise multiplication.

This architecture efficiently expands the model's receptive field exponentially. The GLU acts as a selective filter for the temporal information, while the 1×1 convolution mixes channel features. After L blocks, the final output $\mathbf{H}_L^{(m,t)}$ aggregates high-level temporal features from a large window (length $k\sum_{l=0}^{L-1} 2^l$), providing a rich representation for the subsequent spatial-interaction module.

4.2.2. Dynamic graph construction

A naive implementation would allocate an independent learnable adjacency tensor $\mathcal{A} \in \mathbb{R}^{N_t \times N \times N}$ for each time slot, with N_t denoting the number of daily time slots, but this is both memory-inefficient and fails to exploit low-rank structure. To address this issue, we propose a compact Tucker-style factorization framework, as illustrated in Figure 3. Concretely, we introduce a time-slot embedding $\mathbf{F}^t \in \mathbb{R}^{N_t \times d}$, a source-node

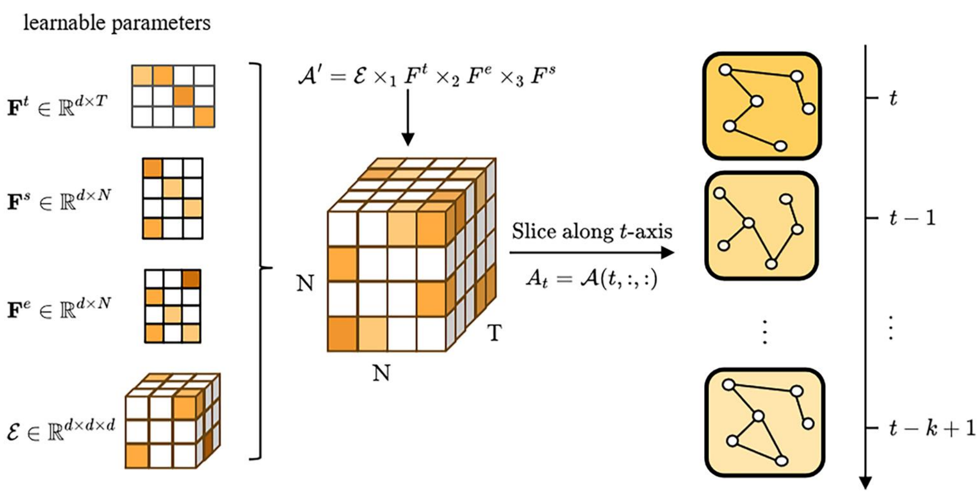

Figure 3. Dynamic graph construction via Tucker decomposition. The spatiotemporal dependency tensor is factorized into a compact core tensor $\mathcal{E}$ and three factor matrices (F^t, F^e, F^s). These components reconstruct the full dependency tensor $\mathcal{A}$, where each time slice $\mathcal{A}_t$ captures evolving spatial relationships.

embedding $\mathbf{F}^e \in \mathbb{R}^{N\times d}$, a target-node embedding $\mathbf{F}^s \in \mathbb{R}^{N\times d}$, and a compact core tensor $\mathcal{E} \in \mathbb{R}^{d\times d\times d}$. The reconstructed tensor is

$$\begin{aligned}\mathcal{A}' &= \mathcal{E}\times_1\mathbf{F}^t\times_2\mathbf{F}^e\times_3\mathbf{F}^s,\\ \mathcal{A} &= \text{Softmax}(\text{ReLU}(\mathcal{A}')),\end{aligned} \tag{5}$$

which yields a distinct adjacency matrix A_t for each time slot by slicing along the first (time) mode.

Importantly $\mathcal{E}, \mathbf{F}^t, \mathbf{F}^e, \mathbf{F}^s$ are learnable model parameters: we randomize them and update them jointly with the rest of the network in back-propagation driven by the observed trip data and the model loss.As a result, the learned directed edge weights are data-driven and adapt to temporal changes in transport system (for example, amplifying links to/from commuting hubs during peak hours), rather than being fixed by physical route maps. This compact Tucker factorization reduces parameter cost from $O(N_t \cdot N^2)$ to $O(d(N_t + 2N) + d^3)$, while enabling expressive, time-varying intramode graphs.

4.2.3. Dynamic graph convolution

Graph convolutions aggregate neighboring information to model spatial dependencies. Following diffusion-based formulations (Lv *et al.* 2018), we treat flow propagation as a diffusion process over the dynamic graph $A^m_{\phi(t)}$. The hidden state update is computed as

$$H^m = \sum_{k=0}^{K}\left(A^m_{\phi(t)}\right)^k H_l^{(m,t)} W, \tag{6}$$

where $H_l^{(m,t)}$ denotes the output hidden states acquired from the temporal convolution layer at block l, W is the learnable weight matrix associated with the k-th diffusion step, and K represents the maximum number of diffusion steps. This operation dynamically aggregates multi-hop spatial information under time-varying connectivity.

4.3. Intermodal interaction

Standard cross-attention mechanisms are widely used to model cross-modal dependencies (Lee *et al.* 2023, Miller *et al.* 2024). However, the full dot-product attention inherently computes all pairwise similarities between queries and keys, which may introduce noisy or irrelevant links across distant regions, thereby weakening predictive focus.

To address this, we propose the SCMI module, which adaptively captures meaningful intermodal influences while suppressing redundant correlations (Figure 4). Specifically, given the hidden representations of a target mode $\mathbf{H}^{m^*} \in \mathbb{R}^{T\times N\times d}$ and exogenous modes $\mathbf{H}^m \in \mathbb{R}^{T\times N\times d}$ ($m \in \mathcal{M}^-$), we compute the query, key, and value matrices as:

$$\begin{aligned}\mathbf{Q}^{(m)} &= \mathbf{H}^{(m)}_{t::}\ \mathbf{W}_Q, \qquad \forall m \in \mathcal{M}^-,\\ \mathbf{K}^{(m^*)} &= \mathbf{H}^{(m^*)}_{t::}\ \mathbf{W}_K,\\ \mathbf{V}^{(m^*)} &= \mathbf{H}^{(m^*)}_{t::}\ \mathbf{W}_V\end{aligned} \tag{7}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d\times d'}$ are learnable parameters.

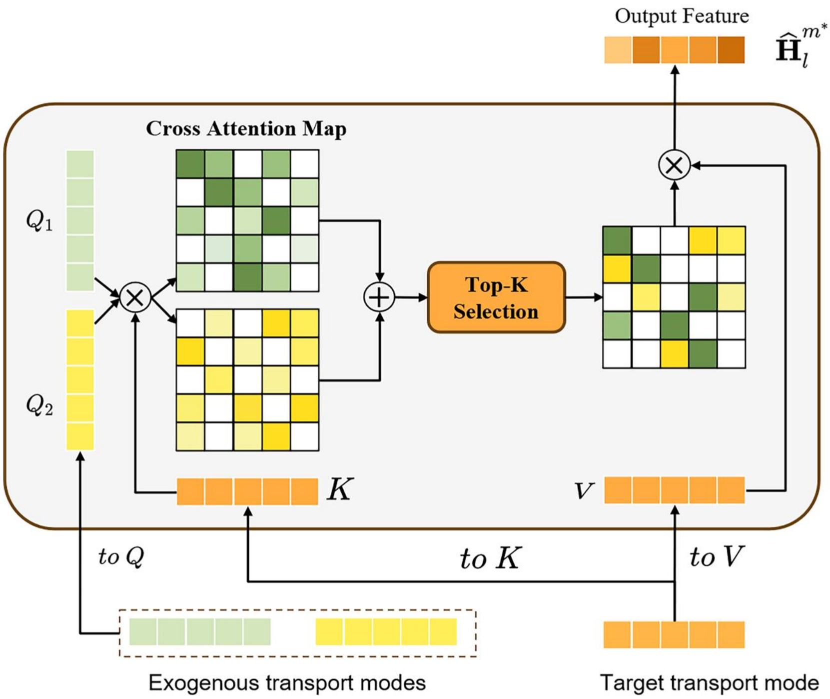

Figure 4. The SCMI module computes sparse cross-attention between a target transport mode and multiple exogenous modes. Queries from exogenous modes interact with keys and values from the target mode, while a top-k operator enforces sparsity to retain only the most significant intermodal relations.

The cross-modal affinities are then computed via dot-products between queries and keys from all exogenous modes and aggregated into a mixed attention map $\mathbf{M}^{\text{mix}} \in \mathbb{R}^{N \times N}$. A *top-k sparsification* operator $\mathcal{T}k(\cdot)$ retains only the strongest correlations, defined as:

$$[\mathcal{T}_k(\mathbf{S})]_{ij} = \begin{cases} S_{ij}, \text{if } S_{ij} \in \text{top-}k(\text{row } i), \\ -\infty, \text{otherwise}. \end{cases} \tag{8}$$

After normalization via Softmax, the refined attention weights are applied to the value matrices across H heads and linearly projected to yield:

$$\hat{\mathbf{H}}^{m^*} = \left[\text{softmax}\left(\mathcal{T}_k \left(\bigoplus_{m \in \mathcal{M}^-} \frac{\mathbf{Q}_h^{(m)} (\mathbf{K}_h^{(m^*)})^T}{\sqrt{d_k}} \right) \right) \mathbf{V}_h^{(m^*)} \right]_{h=1}^{H} \mathbf{W}_O \tag{9}$$

where $[\cdot]$ denotes the concatenation operation implemented across different heads, $\oplus$ represents elementwise aggregation performed across exogenous modes.

By dynamically enforcing sparsity, SCMI concentrates on the most informative region–mode relations, effectively enhancing intermodal spatial modeling and mitigating noise from weak or spurious dependencies.

4.4. Adaptive multitask learning

4.4.1. Output layer

We attach an identity skip connection after every SCMI block. For each mode $m \in \mathcal{M}$ the l-th block outputs $\hat{H}_l^m \in \mathbb{R}^{T \times N \times d_l}$ ($l = 1, \ldots, L$). The residual update step is

$$\tilde{H}_l^m = \hat{H}_l^m + \mathrm{Conv}_{1\times1}\left(\hat{H}_{l-1}^m\right), \qquad l \geq 2, \tag{10}$$

where the 1×1 convolution projects $d_{l-1} \to d_l$, ensuring that the dimensions match for addition purposes.

Then, we apply two mode-specific fully connected layers to produce a τ-step forecast:

$$\hat{Y} = \mathrm{FC}\left(\sigma\left(\mathrm{FC}\left(\mathrm{LN}\left(\tilde{H}_l^m\right)\right)\right)\right), \qquad \hat{Y}^m \in \mathbb{R}^{\tau \times N}, \tag{11}$$

where $\sigma(\cdot)$ is the rectified linear unit (ReLU) function, and $\mathrm{LN}(\cdot)$ denotes the layer normalization operation executed along the channel dimension. The first dense layer enhances the capacity of the model; the second layer maps the latent features to mode-specific τ-step flow prediction.

4.4.2. Uncertainty weighting

We frame multimodal flow forecasting as a multitask learning problem, where predicting the flow for each transport mode constitutes a distinct task. To balance tasks with differing difficulty levels, we adopt uncertainty weighting (Kendall *et al.* 2018). Introducing a learnable homoscedastic uncertainty term σ_m for each mode m, the total loss is

$$\mathcal{L} = \sum \_m \in \mathcal{M}\left[\frac{1}{2\sigma_m^2}\ \mathcal{L}_m + \log \sigma_m\right], \tag{12}$$

where the per-mode loss is

$$\mathcal{L}_m = \frac{1}{\tau N}\sum_{k=1}^{\tau}\sum_{i=1}^{N}|\hat{y}_{i,k}^m - y_{i,k}^m|. \tag{13}$$

The $\log \sigma_m$ term regularizes the optimization process and prevents the acquisition of the trivial solution $\sigma_m \to 0$.

This adaptive formulation enables the model to automatically calibrate the relative importance of each task on the basis of its difficulty and inherent noise. By minimizing the total loss, the model jointly optimizes the predictions obtained for all modes while mitigating the risk of overfitting to tasks with larger or noisier losses. As a result, this strategy facilitates a more robust and balanced multimodal forecasting process within our unified model.

5. Experiments

5.1. Datasets

This study utilizes two large-scale multimodal transportation datasets to evaluate the proposed model: one from Shenzhen (SZ) and the other from New York City (NYC).

Table 2. Summary and comparison of the SZ and NYC datasets.

Category	SZ Dataset	NYC Dataset
Location	Shenzhen (491 TAZs)	New York City (69 TAZs)
Time span	2018/12/01–2019/02/01	2022/12/01–2023/05/31
Time slot	10 minutes	60 minutes
Timestep	8928	4368
Transport modes	Bus,Metro,Taxi	Bus,Metro,Taxi/Ride-hailing
Approx. daily volume	Buses:≈2.85M; Metros:≈1.84M; Taxis:≈0.70M	Buses:≈0.05M; Metros:≈1.67M; Taxis:≈0.1M
Train/Val/Test ratio	7:1:2	7:1:2
Input length	12	12
Output length	12	12

A detailed comparison of their key characteristics is presented in Table 2. The SZ dataset covers December 2018 to February 2019, with 10-minute temporal resolution and three transport modes (bus, metro, and taxi). Bus and metro flows are derived from smart card and GPS data, while taxi flows are reconstructed from trajectory data (Tu *et al.* 2018). The NYC dataset spans December 2022 to May 2023 and includes hourly regional flows for bus[1], metro[2], and taxi[3]. All data originate from official open sources (MTA and TLC) and are aggregated by TAZ.

To prevent temporal data leakage, we partitioned the dataset chronologically into training, validation, and test sets in a 7:1:2 ratio. Crucially, all preprocessing models, such as data scalers, were fitted only on the training set and then used to transform the validation and test sets.In addition, we use the data from 12 steps to predict the flow for the next 12 steps, i.e., a multistep prediction process.

5.2. Experimental settings

5.2.1. Network training

All the experiments are conducted on a machine with an NVIDIA RTX A6000 GPU. We use Python 3.9.13, PyTorch 1.13.0 and BasicTS (Liang *et al.* 2023) platforms to run our models. The hidden dimensionality is 32, and the depth of the encoder layers L is 8. The dimensionality of the periodic time embeddings is 32. For dynamic graph learning, we set all the embedding dimensions to 32. For the SCMI module, the attention dimensionality is 256, with 8 heads. The optimal model is selected according to the performance achieved on the validation set. We train our model via the adaptive moment estimation (Adam) optimizer with a learning rate of 0.001 and weight decay to 0.0003. The model is trained with a batch size of 32 for up to 500 epochs, utilizing an early stopping criterion based on the attained validation performance with a patience of 20 epochs.

5.2.2. Baseline models

We compare the proposed framework against both single modal and multimodal forecasting baselines. For single modal prediction, we include representative models from four categories: MLP-based (STID, MTS-Mixers), GNN-based (STGCN, GWNET), RNN-based (DCRNN, DGCRN), and Transformer-based (Informer, iTransform). For multimodal forecasting, we select two state-of-the-art methods, M2-Former and ST-MRGNN, which jointly predict multimodal urban flow within unified spatiotemporal frameworks. All baselines are trained according to their original configurations; single modal models are trained separately for each mode, while multimodal models generate joint predictions across all modes.

- **STID**: STID tags every variable with a learnable spatial ID embedding and every timestamp with a learnable temporal ID embedding; it concatenates these two IDs with the raw value vector for that time step and maps the result through a small MLP to forecast the next horizon (Shao *et al.* 2022).
- **MTS-Mixers**: MTS-Mixers forecasts multivariate sequences using two factorized MLP mixers: a temporal mixer that linearly mixes the past values of each variable along the time axis and a channel mixer that (via low-rank factorization) mixes the variables within each timestep; stacking these mixers and finishing with a simple linear head yields the future horizon (Li *et al.* 2025).
- **STGCN**: This is a GCN-based deep learning model that captures spatiotemporal dependencies through spatial graph convolutional layers and temporally gated causal convolutional layers (Yu *et al.* 2018).
- **GWNET**: This is a spatiotemporal graph learning approach that employs node embeddings to model complex spatial dependencies and utilizes dilated causal convolutions for temporal modeling purposes (Wu *et al.* 2019).
- **DCRNN**: This model learns spatiotemporal patterns within graph-structured data by integrating graph diffusion convolution to capture spatial dependencies and RNNs to model temporal dynamics (Lv *et al.* 2018).
- **DGCRN**: Built upon an RNN model, this model captures dynamic spatiotemporal patterns by generating dynamic node embeddings and time-varying adjacency matrices at each step, which are fused with a static graph within a specialized recurrent GCN module (Li *et al.* 2021b).
- **Informer**: This is an efficient transformer-based model for long-sequence time series prediction that leverages a multihead ProbSparse self-attention mechanism and a distillation architecture to increase its scalability and performance (Zhou *et al.* 2021).
- **iTransform**: This model restructures the transformer architecture for conducting multivariate time series forecasting by treating the entire time series of each variable as a token. This inversion scheme allows the model to better capture intervariable correlations through its self-attention mechanism (Liu *et al.* 2024).
- **M2-Former**: A multitask Transformer-based model for multimodal inflow prediction, which adopts an encoder-decoder structure. The encoder combines temporal attention with adaptive multigraph convolution to capture dynamic crossmodal dependencies, while the decoder extracts mode-specific features and supports crossmode knowledge sharing for joint urban flow forecasting (Yang *et al.* 2024).
- **ST-MRGNN**: It constructs multirelational spatiotemporal relation graphs to capture heterogeneous spatial dependencies, integrates them with gated temporal convolutions, and employs an attention-based aggregation mechanism to model crossmodal spatiotemporal correlations for joint forecasting (Ke *et al.* 2021).

5.2.3. Evaluation metrics

To evaluate the performance of the tested models for each transport mode, we adopt the mean absolute error (MAE), root mean square error (RMSE), and the coefficient of determination (R^2). For transport mode m, these metrics are defined as:

$$\mathrm{MAE}_m = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|, \tag{14}$$

$$\mathrm{RMSE}_m = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}, \tag{15}$$

$$R_m^2 = 1 - \frac{\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}, \tag{16}$$

where y_i and $\hat{y}_i$ denote the ground-truth and predicted values of transport mode m at region i, respectively, and $\bar{y}$ is the mean of the ground-truth values. When evaluating the metro mode, computation is limited to regions containing metro stations using a binary mask $\mathbf{M}$, where $M_i = 1$ if region i has a station and $M_i = 0$ otherwise. Metrics are calculated only for $M_i = 1$ regions to ensure metro-specific evaluation.

5.3. Performance evaluation

5.3.1. Overall performance across cities

Overall performance on Shenzhen and New York City datasets is summarized in Tables 3 and 4. Each metric represents the mean across 12 forecast horizons. MM-DyGNN consistently achieves the best results across all modes and cities, outperforming both single-modal and multimodal baselines. On the SZ dataset, it yields the lowest MAE for buses (7.68, 7.5% gain), metros (18.76, 6.9% gain), and taxis (2.17, 1.2% gain). Similar gains are observed for NYC—buses (9.27, 8.8%), metros (153.86, 5.8%), and taxis (10.36, 4.9%). These results confirm that integrating cross-modal information with dynamic graph learning substantially enhances forecasting accuracy across diverse urban contexts.

Table 3. Overall performance comparison on the SZ dataset.

	Buses			Metros			Taxis		
Model	MAE ↓	RMSE ↓	R^2 ↑	MAE ↓	RMSE ↓	R^2 ↑	MAE ↓	RMSE ↓	R^2 ↑
STID	13.2422	35.0326	0.7153	38.9733	105.6388	0.5922	3.0743	7.3313	0.8738
MTS-Mixers	10.3709	26.0494	0.8426	34.5109	109.0358	0.5656	2.6471	6.2272	0.9089
GWNet	8.3475	19.6434	0.9105	20.1525	57.3505	0.8798	2.1965	4.8448	0.9449
STGCN	9.0046	21.4736	0.8930	25.6273	69.7929	0.8220	2.4574	5.5091	0.9287
Informer	8.2969	19.0810	0.9155	24.7346	65.1595	0.8449	2.3083	5.1405	0.9379
iTransformer	11.0848	28.3017	0.8141	28.9338	83.5492	0.7449	2.6883	6.2116	0.9094
DCRNN	10.1547	28.2374	0.8150	31.0589	92.9277	0.6844	2.3401	5.2962	0.9341
DGCRN	8.5699	21.4108	0.8936	20.5376	60.0824	0.8681	2.1936	4.8387	0.9450
M2-Former	8.8824	20.1866	0.9055	21.1998	67.3503	0.8341	2.5196	5.4649	0.9299
ST-MRGNN	9.6416	23.0301	0.8769	27.5702	77.8338	0.7784	2.5937	6.0196	0.9149
MM-DyGNN	**7.6768**	**16.8845**	**0.9334**	**18.7602**	**51.9242**	**0.9014**	**2.1673**	**4.6263**	**0.9497**

Notes: **Bold** and underlined values denote the best and second-best results, respectively. Arrows (↓/↑) indicate whether lower or higher is better.

Table 4. Overall forecasting performance on the NYC dataset.

Model	Buses MAE ↓	Buses RMSE ↓	Buses R^2 ↑	Metros MAE ↓	Metros RMSE ↓	Metros R^2 ↑	Taxis MAE ↓	Taxis RMSE ↓	Taxis R^2 ↑
STID	10.6354	18.7508	0.8584	225.3718	310.3428	0.9321	10.8969	18.8311	0.9501
MTS-Mixers	16.7447	28.6899	0.6685	308.1846	513.2314	0.8143	18.7112	29.5210	0.8780
GWNet	10.1689	17.8218	0.8721	163.2514	289.0537	0.9411	10.9577	18.9613	0.9493
STGCN	11.1954	19.9843	0.8391	182.5952	323.1724	0.9263	12.5890	21.2686	0.9362
Informer	11.1534	18.7767	0.8580	175.5140	306.7315	0.9336	11.2088	19.3430	0.9471
iTransformer	11.3044	21.2856	0.8175	204.5588	336.9467	0.9199	14.7790	24.7386	0.9144
DCRNN	11.3848	21.3816	0.8159	194.9248	329.2442	0.9236	14.1558	24.1540	0.9184
DGCRN	19.3546	34.4659	0.5215	197.3286	330.3192	0.9231	11.2934	19.5609	0.9460
M2-Former	10.4326	18.0181	0.8692	170.0826	298.3192	0.9372	11.8233	19.6685	0.9454
ST-MRGNN	10.1740	18.4338	0.8631	176.5605	316.1079	0.9295	13.0808	21.3690	0.9356
MM-DyGNN	**9.2743**	**16.5996**	**0.8890**	**153.8571**	**281.7863**	**0.9440**	**10.3606**	**18.2947**	**0.9528**

Notes: **Bold** and underlined values denote the best and second-best results, respectively. Arrows (↓/↑) indicate whether lower or higher is better.

Table 5. Forecasting performance on SZ dataset across prediction horizons (bus mode shown as a representative example).

Transportation	Model	H = 3 MAE ↓	H = 3 RMSE ↓	H = 3 R^2 ↑	H = 6 MAE ↓	H = 6 RMSE ↓	H = 6 R^2 ↑	H = 12 MAE ↓	H = 12 RMSE ↓	H = 12 R^2 ↑
Buses	STID	9.3297	21.2441	0.8956	12.8352	32.1086	0.7610	18.6849	49.4051	0.4308
	MTS-Mix	7.9512	17.3180	0.9306	9.8799	23.2258	0.8750	14.2258	37.8541	0.6659
	GWNet	7.1340	15.4203	0.9449	8.2097	19.2107	0.9144	10.1667	24.9143	0.8552
	STGCN	7.6647	16.6965	0.9355	8.7340	20.4378	0.9032	11.1769	27.9782	0.8175
	Informer	7.3958	15.9976	0.9408	8.1475	18.3379	0.9220	9.6196	23.5952	0.8702
	iTransformer	8.0555	18.0277	0.9245	10.3574	26.0011	0.8433	14.6251	39.1962	0.6418
	DCRNN	7.6220	16.9969	0.9331	9.7954	25.2283	0.8525	13.8176	40.7731	0.6123
	DGCRN	6.8449	14.2979	0.9525	8.5257	20.2824	0.9047	10.4398	27.6744	0.8215
	M2-Former	7.3329	15.1743	0.9467	8.6642	18.8478	0.9177	11.5084	27.6795	0.8213
	ST-MRGNN	7.6295	16.7622	0.9347	9.3040	21.8536	0.8893	12.5641	30.2858	0.7861
	MM-DyGNN	**6.7888**	**13.5315**	**0.9574**	**7.6052**	**15.8744**	**0.9415**	**9.4851**	**20.8994**	**0.8981**

Notes: **Bold** and underlined values denote the best and second-best results, respectively. Arrows (↓/↑) indicate whether lower or higher is better.

Models lacking spatial structure, such as STID and MTS-Mixers, perform markedly worse; their metro MAE on SZ (38.97, 34.51) and NYC (225.37, 308.18) exceed MM-DyGNN's by over 80–100%. While GNN-based models (STGCN, GWNet) capture spatial patterns, they remain constrained by static graphs. MM-DyGNN improves MAE by up to 8.0% on SZ and 8.8% on NYC. Existing multimodal methods (M2-Former, ST-MRGNN) also underperform due to fixed or noisy intermodal links. In contrast, MM-DyGNN's adaptive dynamic graph and sparse cross-modal interaction mechanisms effectively filter irrelevant dependencies and amplify critical intermodal signals, ensuring more accurate and robust urban flow forecasting.

5.3.2. Performance evaluation across prediction horizons

By examining results at multiple horizons, we can assess the model's responsiveness to short-term demand fluctuations as well as its capacity to capture long-range temporal dependencies. Table 5 presents the forecasting performance of different models for the bus mode in the SZ dataset at three representative horizons ($H = 3, 6, 12$). Overall, MM-DyGNN consistently outperforms all baselines, achieving the lowest MAE and RMSE and the highest R^2 across all horizons. For instance, at the long-term horizon ($H = 12$), MM-DyGNN attains an RMSE of 20.90, which is substantially lower than the second-best model (Informer, 23.60), reflecting superior temporal stability and error control. The RMSE of MM-DyGNN increases moderately from 13.53 to 20.90 as the horizon extends from $H = 3$ to $H = 12$, indicating effective mitigation of cumulative error propagation over time. These results demonstrate that MM-DyGNN captures both short-term variations and long-term dependencies in multimodal traffic flows more effectively than existing GNN- and Transformer-based baselines.

5.4. Contribution of the intramodal dynamic graph constructor

Modeling dynamic spatial dependencies—shaped by time-varying factors like congestion, incidents, and commuting patterns is a key challenge in predicting urban multimodal flows. To address this, our dynamic graph learning module employs time-specific graphs to capture these evolving relations. We assess its effectiveness in an ablation study by replacing the dynamic graph with two alternatives:

- **Time-invariant adaptive graph**: We collapse the time axis ($N_t = 1$) and learn a single adjacency matrix per mode that is shared across all horizons and time slots. The graph weights are still learned from data (adaptive), but do not vary over time.
- **Predefined spatial graph**: In this variant, the learned intramodal dynamic graph is replaced with a predefined, static adjacency matrix constructed based on the spatial contiguity of TAZs.

As summarized in Table 6, the dynamic graph variant (MM-DyGNN) consistently outperforms both alternatives across all transport modes. Compared to the time-invariant adaptive graph, it achieves MAE improvements of 4.76% (Bus), 3.74% (Metro), and 2.33% (Taxi). The predefined spatial graph performed worst, increasing MAE by 16.72% (Bus) and 5.62% (Taxi) relative to the full dynamic model. These results confirm that modeling temporally dynamic spatial dependencies is essential for enhancing predictive accuracy.

Figure 5 provides a more intuitive demonstration of how the dynamic graph constructor captures the temporal evolution of spatial dependencies. During the late-night period (00:00), the learned adjacency matrix exhibits weak and diffuse connections, suggesting limited travel interactions across TAZs. By the midday period (11:00), connectivity intensifies and several TAZs begin to form local hubs, indicating moderate urban mobility. The most pronounced transformation occurs during the morning

Table 6. Performance comparison results obtained while considering different graph construction strategies.

	Buses			Metros			Taxis		
Model Variant	MAE ↓	RMSE ↓	R^2 ↑	MAE ↓	RMSE ↓	R^2 ↑	MAE ↓	RMSE ↓	R^2 ↑
MM-DyGNN	**7.6768**	**16.8845**	**0.9338**	**18.7602**	**51.9242**	**0.9014**	**2.1673**	**4.6263**	**0.9497**
Time-invariant adaptive graph	8.0605	18.1784	0.9233	19.4894	52.3267	0.8999	2.2190	4.8960	0.9437
Predefined spatial graph	8.9605	20.5423	0.9021	19.8713	53.5341	0.8952	2.2890	4.9641	0.9421

Notes: Bold values denote the best results. Arrows (↓/↑) indicate whether lower or higher values are better.

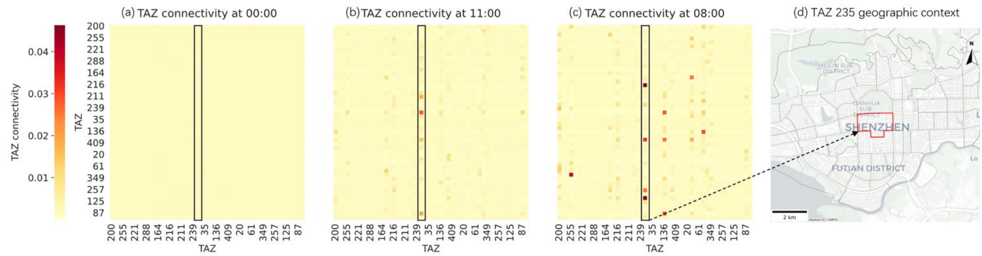

Figure 5. Temporal evolution of interregional bus connectivity around TAZ 235 (Shenzhen Civic Center). The learned adjacency matrices illustrate weak and diffuse connections at 00:00 (late-night), moderate clustering at 11:00 (midday), and strong flow at 08:00 (morning peak).

commuter peak (08:00), where connectivity values surge sharply around TAZ 235—the Shenzhen Civic Center—forming a strong bidirectional flow network that reflects the spatial concentration of commuting demand. Overall, the visualization reveals that the proposed dynamic graph constructor effectively captures both temporal transitions and spatial concentration patterns of urban flow, aligning well with real-world commuting dynamics.

5.5. Contribution of the sparse cross-modal interaction module

The SCMI module captures interconnections among multiple transport modes across regions, enabling the model to extract key cross-modal dependencies. To evaluate its contribution, we conduct ablation studies using three MM-DyGNN variants:

- **w/o Top-*k* Selection Operation**: In this variant, the top-*k* selection operation is removed, eliminating the sparsity constraint. The model instead applies standard full cross-attention, potentially introducing noise from irrelevant regions.
- **w/o Attention Interaction**: In this variant, the cross-modal attention mechanism is entirely removed and replaced by direct feature summation, simply adding exogenous modal features to the target mode without attention-guided weighting.
- **w/o SCMI**: In this most restrictive setting, the entire SCMI module is omitted. Each transport mode is modeled independently, and features remain isolated without exploiting intermodal complementarities or substitutions.

Table 7 shows that incorporating sparse cross-attention consistently improves performance across all modes. Relative to the full attention variant, MM-DyGNN reduces MAE by 4.4% and RMSE by 6.6% for buses, with similar but smaller gains for metros and taxis. Removing attention interaction further degrades performance (e.g., bus MAE +7.3%), confirming the necessity of attention-guided weighting. The largest drop occurs when SCMI is fully excluded, especially for metros (+33.4% MAE), highlighting the strong intermodal influence. Overall, the study demonstrates that cross-modal attention drives most of the gains, while top-*k* sparsification further enhances signal-to-noise ratio and prediction accuracy.

Table 7. Performance comparison results obtained while considering cross-modal interactions.

	Buses			Metros			Taxis		
Model Variant	MAE↓	RMSE↓	R^2 ↑	MAE↓	RMSE↓	R^2 ↑	MAE↓	RMSE↓	R^2 ↑
MM-DyGNN	**7.6768**	**16.8845**	**0.9338**	**18.7602**	**51.9242**	**0.9014**	**2.1673**	**4.6263**	**0.9497**
w/o Top-*k* Selection Operation	8.0306	18.0760	0.9242	19.4576	52.2071	0.9003	2.2182	4.7337	0.9473
w/o Attention Interaction	8.2376	18.5060	0.9205	20.6751	54.6053	0.8910	2.2792	5.0147	0.9409
w/o SCMI	8.6353	19.8402	0.9086	25.0346	63.8986	0.8508	2.4486	5.2960	0.9341

Notes: Bold values denote the best results. Arrows (↓/↑) indicate whether lower or higher values are better.

Table 8. Performance comparison among variants with different loss functions.

	Buses			Metros			Taxis		
Loss function	MAE↓	RMSE↓	R^2 ↑	MAE↓	RMSE↓	R^2 ↑	MAE↓	RMSE↓	R^2 ↑
Uncertainty-Weighted Loss	**7.6768**	**16.8845**	**0.9338**	**18.7602**	**51.9242**	**0.9014**	**2.1673**	**4.6263**	**0.9497**
MAE Addition	8.7306	18.9760	0.9164	18.7706	51.9371	0.9013	2.1821	4.6837	0.9485

Notes: Bold values denote the best results. Arrows (↓/↑) indicate whether lower or higher values are better.

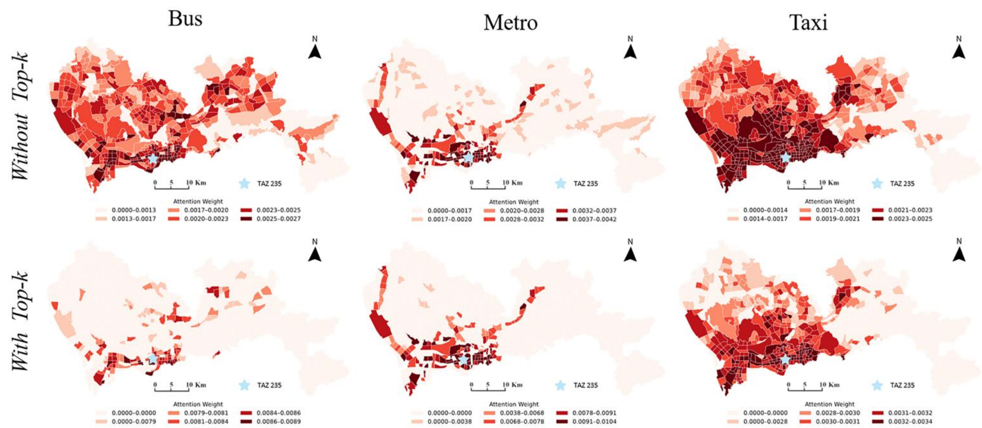

Figure 6. Spatial distribution of the attention weights learned over TAZ 235 (Shenzhen's Civic Center) by the SCMI module. Rows: without top-*k* (top) vs. top-*k* (bottom); Columns: bus-metro-taxi. Darker shades indicate larger attention scores.

Figure 6 compares spatial attention maps from the SCMI module with and without top-*k* sparsification to illustrate how sparsity refines cross-modal dependencies. Focusing on TAZ 235 (Shenzhen Civic Center), the non-sparse setting (top row) produces diffuse and noisy attention, with peripheral TAZs in western and northern Shenzhen receiving non-negligible weights. After applying top-*k* selection (bottom row), attention becomes sharply localized around key east–west transit corridors in southern Shenzhen, aligning with dense metro and bus routes, while taxis maintain broader spatial influence. This sparsification improves MAE by 2.3–4.4% over the full-attention variant (Table 7), confirming that the top-*k* mechanism effectively removes spurious long-range dependencies and enhances focus on functionally meaningful cross-modal relations.

5.6. Contribution of the uncertainty-weighted loss

To evaluate whether uncertainty modeling can serve as an implicit task-weighting mechanism in our multimodal (buses-metros-taxis) forecasting network, we replace the uncertainty-weighted loss with a plain MAE summation-based variant:

- **MAE Addition**: In this variant, the total loss is defined as the sum of MAE of each transport mode: $\mathcal{L} = \mathcal{L}_{\text{bus}} + \mathcal{L}_{\text{metro}} + \mathcal{L}_{\text{taxi}}$, where each $\mathcal{L}_m$ is computed as the MAE for mode m, analogous to the weighted loss formulation.

As shown in Table 8, compared with plain MAE summation, the uncertainty-weighted loss achieves a notable 12.1% reduction in bus MAE (from 8.73 to7.68) and a slight improvement for taxis, while metro performance remains stable. This demonstrates that uncertainty weighting effectively balances task learning and enhances overall multimodal forecasting accuracy.

6. Discussion

6.1. Robustness under spatial occlusion

We evaluate the robustness of MM-DyGNN by randomly masking 10%, 20%, and 30% of spatial nodes and measuring performance degradation. Two strong single-modal baselines, GWNET and DGCRN, are used for comparison. As shown in Figure 7, MM-DyGNN consistently exhibits the smallest performance drops across all modes and masking ratios. When 30% of nodes are masked, its MAE for bus flow increases by only 7.87, compared to 10.63 for GWNET and 11.21 for DGCRN, corresponding to 26–30% higher robustness. The advantage is more evident in metro prediction, where MM-DyGNN's degradation (19.36%) is much lower than GWNET (47.30%) and DGCRN (30.12%), confirming the effectiveness of multimodal modeling in maintaining stability under missing data.

This resilience stems from the model's ability to leverage cross-modal correlations to infer missing patterns. When data from one mode are partially unavailable (e.g., bus sensors at a hub), MM-DyGNN references concurrent metro or taxi flows to

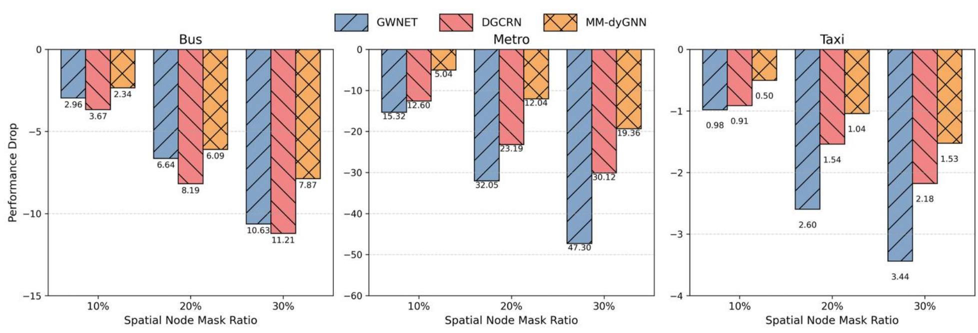

Figure 7. Performance degradation under spatial node masking. Relative MAE increases of GWNET, DGCRN, and MM-DyGNN across masking ratios (10%, 20%, 30%) for bus, metro, and taxi prediction.

recover information. Such multimodal fusion enhances robustness by providing complementary contextual cues and acts as an implicit regularization mechanism against data corruption (Zhang *et al.* 2025). In contrast, single-modal models depend solely on within-mode correlations, making them more vulnerable to localized data loss. The integration of multimodal data thus improves both predictive accuracy and reliability in real-world intelligent transportation systems (Huang *et al.* 2022b).

6.2. Hyperparameter analysis

6.2.1. Sensitivity to the sparsity level

We analyze the effect of sparsity in cross-modal attention by varying the top-k parameter, which controls the number of retained connections in the SCMI module. As shown in Figure 8, smaller k values (high sparsity) may remove informative links and slightly increase MAE, whereas larger k values introduce redundant or noisy dependencies. MAE first decreases with increasing k, reaches an optimal level, and then rises as k approaches full connectivity. The optimal k differs by mode: $k = \frac{N}{4}$ for buses and metros, and $k = \frac{2N}{3}$ for taxis, indicating mode-specific interaction density. Bus and metro flows are more sensitive to k, while taxi performance remains relatively stable. Overall, these results highlight the importance of tuning k to balance expressiveness and sparsity. Properly chosen k values enhance both prediction accuracy and interpretability by filtering out weak cross-modal dependencies and emphasizing meaningful multimodal interactions.

6.2.2. Sensitivity to diffusion steps

We examine the effect of the diffusion step parameter K, which defines the spatial receptive field in the dynamic graph convolution layer (Equation 6). As shown in Figure 9, increasing K expands the range of spatial dependencies but can cause oversmoothing and higher computational costs (Li *et al.* 2018; Chen *et al.* 2020). When $K = 1$, the model captures only immediate neighbors, yielding high RMSE (e.g., 60.74 for metro). Performance improves as K increases and reaches its best at $K = 2$, achieving the lowest RMSE for all modes (bus 16.88, metro 51.92, taxi 4.63). Beyond this point, performance declines as redundant distant information dilutes local patterns. Thus, $K = 2$ provides the optimal trade-off between local precision and global awareness across transport modes.

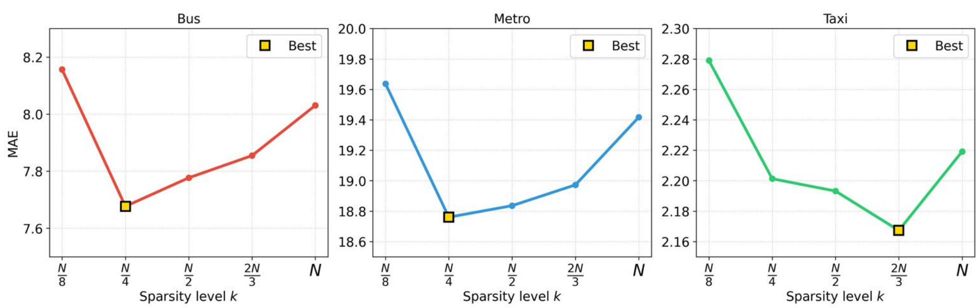

Figure 8. Sensitivity of model performance (MAE) over sparsity level k on SZ dataset across modes. The parameter k is varied over $\{\frac{N}{8}, \frac{N}{4}, \frac{N}{2}, \frac{2N}{3}, N\}$.

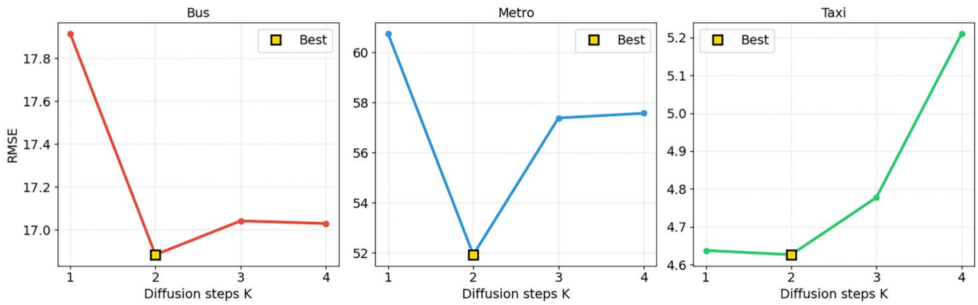

Figure 9. Sensitivity of model performance (RMSE) over diffusion steps *K* on SZ dataset across modes.

7. Conclusion

In this study, we present the MM-DyGNN, a robust predictive framework designed for complex multimodal traffic environments. The model jointly captures spatial, temporal, and cross-modal dependencies through three core innovations: a dynamic graph constructor that models time-varying spatial relations, a sparse cross-modal interaction (SCMI) mechanism that strengthens intermodal information exchange, and an uncertainty-aware multitask loss that balances learning across heterogeneous transport modes. Comprehensive experiments in Shenzhen and New York City datasets demonstrate that MM-DyGNN consistently outperforms state-of-the-art baselines in bus, metro and taxi modes, achieving MAE reductions of 1.2–8.8% and exhibiting up to 30% lower performance degradation under simulated sensor failures. These results highlight the model's ability to deliver accurate and resilient multimodal flow forecasting, underscoring its potential to support data-driven and sustainable urban mobility management.

The main contributions of this study are summarized as follows: (i) We propose a novel multimodal time-varying dynamic graph learning module based on Tucker decomposition, which jointly captures temporal variations in intramodal spatial dependencies and the heterogeneity across different transportation modes. (ii) We develop a sparse cross-modal interaction (SCMI) mechanism that selectively identifies and exploits key cross-modal spatial correlations, thereby enhancing both the efficiency and interpretability of multimodal learning. (iii) We introduce an adaptive multitask learning strategy with uncertainty weighting that dynamically balances the optimization of heterogeneous modal objectives, improving the overall forecasting accuracy and reliability of the model. (iv) We conduct comprehensive cross-city evaluations and visualization analyses using multimodal datasets from Shenzhen and New York City, demonstrating the model's superior accuracy, robustness, and practical interpretability in distinct urban contexts.

Despite the strong performance of MM-DyGNN, several limitations warrant future investigation. (i) The model currently omits external contextual factors (e.g., weather, events) and domain priors such as road networks or land use, which could enhance accuracy and generalization. (ii) Although it captures key cross-modal interactions, it does not yet distinguish their semantic nature (e.g., competition or complementarity). (iii) The top-k sparsification ($k = N/4$) remains heuristic, and future work will explore adaptive or learnable selection mechanisms. (iv) Finally, while computationally feasible, inference efficiency on large-scale datasets is still constrained by the cost of dynamic graph learning and feature fusion.

Notes

1. https://data.ny.gov/Transportation/MTA-Bus-Hourly-Ridership-Beginning-February-2022.
2. https://data.ny.gov/Transportation/MTA-Subway-Hourly-Ridership-2020-2024/wujg-7c2s/about_data.
3. https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page.

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Data availability statement

The source data and codes supporting the findings of this study are available at https://doi.org/10.6084/m9.figshare.29476379.v1.

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